The Generalized-Sturmian Secular Equation Block-Encodes Without a Metric: A Quantum-Algorithm Analysis, Metric-Free for Atoms and a Conditioning Frontier for Molecules*

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(Dated: August 18, 2026)

A recurring intuition in quantum simulation is that a Coulomb–Sturmian / hyperspherical basis is well suited to a quantum computer, because its angular matrix elements are exact rationals generated on the device from quantum-number labels rather than loaded as a classically-computed tensor; only the one-dimensional radial content need be shipped in. We test the load-bearing obstruction to that idea—the cost of *encoding* the basis—and find it turns sharply on how the basis is posed. **[MEASURED]** Encoded naively, in the ordinary L^2 Schrödinger framework, the genuine shared-scale Coulomb–Sturmian basis is L^2-non-orthogonal; a second-quantized (Jordan–Wigner) encoding must Löwdin-orthogonalize it, and the LCU 1-norm λ then *inflates* as $\lambda \sim Q^{3.3}$ (versus $Q^{1.2}$ for an orthonormal hydrogenic basis), driven by the growing ill-conditioning of the overlap. **[ESTABLISHED, from Avery's formulation]** The generalized-Sturmian *isoenergetic* reformulation removes this obstruction for atoms: posed in the potential-weighted metric in which the Coulomb–Sturmians are orthonormal, the atomic secular equation is a *standard* (metric-free) eigenproblem $[\,\mathrm{diag}(ZR_\nu) + T' - p_\kappa \mathbb{1}\,]B = 0$ whose eigenvalues are the scaling parameters $p_\kappa = \sqrt{-2E}$ directly and whose interelectron matrix T' is a matrix of pure numbers, independent of p_κ, of energy, and of Z. **[MEASURED]** We implement it, validate it against helium (a single Goscinskian configuration reproduces the textbook variational value -2.847 Ha; adding configurations lowers the energy monotonically through the s, $+p$ and *spdf* sectors—past the s-sector limit -2.873 Ha to -2.897 Ha, toward the exact non-relativistic -2.90372 Ha), and measure the decisive quantity: the block-encoding 1-norm of the secular matrix grows *sublinearly*, $\|M\|_1 \sim K^{0.84}$ (full s+p+d+f basis; $K^{0.78}$ s-only) in the number of configurations—the opposite of the naive inflation. **[OBSERVATION]** A targeted prior-art search finds no existing quantum algorithm combining a Sturmian basis, a quantum eigenvalue routine, and the isoenergetic inversion; the two documented cost risks for such an algorithm—metric conditioning and an outer energy search—are both *absent* in the atomic case (no metric; the eigenvalues are the energies). **[MEASURED]** For molecules the Shibuya–Wulfman metric returns and the problem is again generalized; we measure its conditioning and find it uniformly better than the L^2 overlap—its intra-center block is exactly the identity—and markedly better at larger internuclear separation, but still growing with basis size through the inter-center coupling. **[SYMBOLIC + MEASURED]** The growth law is derived (a band-limited concentration rate, $1 - \sigma_{\max} \propto (kR)^2/n^2$, giving asymptotic exponent 2—the leading order is derived, the remainder measured rather than bounded; the fitted window exponents 1.85/1.97 are pre-asymptotic readings). **[MEASURED]** The same cross-center singular spectrum carries the composition-wall commutator, whose norm saturates at $\frac{1}{2}$. **[RESOURCE MODEL]** Turning this into a block-encoding count for H_2^+ (validated to bind, 1.3% s-only), the payoff is the gerade/ungerade lever: the σ_g ground state lives in the gerade sector, whose conditioning is *flat*—and flat at an exact constant of the sinc symbol, $2/(1 + \min_x j_0(x)) = 2.555041\ldots$, independent of R and of basis size—so the ground-state metric penalty does *not* grow with basis size—a $\sim$60$\times$ reduction versus the full metric and 10^3–$10^4\times$ smaller than a standard (ratio-dependent) Gaussian metric for the same problem, at $\sim$30 qubits. **[MEASURED]** The penalty also does not compound with electron number: the metric is discharged once at the one-electron molecular-orbital construction, after which the many-electron secular equation carries an identity metric—the metric cost is one-electron, not N-electron. **[OPEN]** The atomic case is a clean, apparently novel, metric-free quantum secular equation; the molecular case is a genuine frontier in which the metric cost is mitigated for excited/ungerade states, essentially removed for the ground state of *equivalent-center* systems (a water probe shows a symmetry-inequivalent heavy center reinstates it), and confined to one electron. **[MEASURED]** Its remaining question—whether the N-electron 1-norm stays sublinear as it does for atoms—we answer in the negative: the atomic sublinearity rides on a diagonal nuclear matrix $T^0 = ZR_\nu$ that is single-center-specific, and a validated two-center CI (dissociating H_2 correctly) has a *standard* block-encoding 1-norm growing polynomially ($\sim n_{\mathrm{orb}}^{2.2}$), with no sublinear behavior. The real molecular savings are the metric levers (gerade sector, large separation, one-electron-only metric), not a sublinear matrix. We do not demonstrate the algorithm on hardware; this is a structural and resource analysis.

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* Paper 60 in the Geometric Vacuum series

I. INTRODUCTION

The generalized-Sturmian method of Avery and Avery [1–4], built on Goscinski's weighted wave equation and the momentum-space Fock projection, is one of the most elegant classical frameworks for atomic and molecular electronic structure. Its angular content is closed-form—Gaunt integrals, Wigner $3j/6j$ symbols, Shibuya–Wulfman integrals—generated from the quantum numbers (n, ℓ, m) with no spatial quadrature. This is the same discrete angular skeleton that the Geometric Vacuum program reads as a graph (Papers [15, 16]); the two frameworks meet exactly at those closed-form angular integrals.

The angular cleanness suggests a quantum-computing use, articulated independently in the Sturmian and Geometric-Vacuum settings: the bottleneck in quantum simulation is often not the term count of the Hamiltonian but the *input/output*—classically computing a two-electron integral tensor and loading it onto the device. If the angular structure can be generated on the device from labels, and only the one-dimensional radial factors loaded, the loading cost is dominated by a small radial-seed count rather than an $O(M^4)$ tensor. This is a known-good lever in fault-tolerant simulation: plane-wave quantum-simulation methods compute their coefficients rather than loading them and win decisively for large basis sets [7, 8] (first quantization, with $O(\eta \log N)$ qubits, at the [8] end).

The lever has a well-documented Achilles heel, however. Its payoff is governed not by the loading count alone but by the *LCU* 1*-norm* $\lambda = \sum_i |c_i|$ of the encoded Hamiltonian, which sets the query and T-gate cost of qubitization and block-encoding [9, 10]; plane waves win the I/O yet pay a large λ. The question this paper addresses is whether a Coulomb–Sturmian encoding keeps λ bounded. We find that the answer depends entirely on *how the basis is posed*, and that the isoenergetic reformulation—the defining move of the generalized-Sturmian method—is precisely what makes the atomic problem cheap.

The paper is organized as the argument runs. Section II measures the naive obstruction. Section III states the isoenergetic reformulation and why it is metric-free for atoms. Section IV implements and validates it on helium and measures the sublinear 1-norm. Section V states the quantum algorithm and its novelty. Section VI measures the molecular Shibuya–Wulfman conditioning. Section VII states the honest boundary.

II. THE NAIVE OBSTRUCTION: 1-NORM INFLATION FROM L^2 ORTHOGONALIZATION

A second-quantized (Jordan–Wigner) encoding requires an L^2-orthonormal set of single-particle spin-orbitals. The genuine shared-scale Coulomb–Sturmians are *not* L^2-orthonormal: at a common scale k their higher members grow linearly dependent, so a qubit encoding must first orthogonalize them (Löwdin, $S^{-1/2}$), and the coefficient distribution—hence $\lambda = \sum_i |c_i|$—spreads.

[MEASURED] In a controlled s-sector helium construction in which only the radial convention is changed and the ground shell is anchored so the two families coincide at $Q = 2$ qubits, the LCU 1-norm scales as

$$\lambda_{\text{hydrogenic}} \sim Q^{1.19}, \qquad \lambda_{\text{Sturmian}} \sim Q^{3.33}, \tag{1}$$

a factor of ~ 18 larger by ten qubits (a measured ratio from the $\lambda(Q)$ data; like Eq. (5) these exponents are numerical fits, and the tracked claim is the sublinear-vs-superlinear contrast, not the precise value). The mechanism is the L^2 overlap S: with helium Coulomb–Sturmians its condition number climbs $(3.0, 5.8, 13.9, 32.2$ for $N = 2, 3, 5, 8)$, and the dense $S^{-1/2}$ built from an ill-conditioned Gram inflates λ. This is a currency-change of the familiar cost-conservation phenomenon in non-orthogonal quantum chemistry (the overlap metric never disappears; moving it densifies operators, states, or coefficients [15]). Encoded this way, the Sturmian basis is the *worst*, not the best, choice.

[MEASURED] The obstruction is intrinsic to the L^2 posing, not to a choice of orthogonalization: re-solving the converged $s+p+d+f$ helium problem of Sec. IV as the generalized eigenproblem $MB = pSB$ with the L^2 overlap S restored *diverges*, the metric condition number climbing $4 \to 3673$ over the basis while the energy runs to -3×10^6 Ha—the metric-free standard eigenproblem of Sec. III is not merely convenient, it is the only well-conditioned posing we have found. The rest of the paper shows this obstruction is an artifact of the L^2 framing, not of the Sturmians.

III. THE ISOENERGETIC REFORMULATION IS METRIC-FREE FOR ATOMS

The Coulomb–Sturmians obey a *potential-weighted* orthonormality relation [3, 5, 6],

$$\int d^3x \, \varphi^*_{n'\ell'm'}(x) \, \frac{1}{r} \, \varphi_{n\ell m}(x) = \frac{k}{n} \, \delta_{n'n}\delta_{\ell'\ell}\delta_{m'm}, \tag{2}$$

orthonormal in the $1/r$ weight, *not* in L^2. **[MEASURED]** We confirm Eq. (2) numerically: the potential-weighted overlap is diagonal to $\sim 10^{-6}$ off-diagonal, whereas the L^2 overlap of the same basis is the ill-conditioned matrix of Sec. II.

Building the electronic-structure problem in this native metric is the generalized-Sturmian method. With the basis potential taken to be the bare nuclear attraction $V_0 = -\sum_j Z/r_j$, the Goscinskian configurations Φ_ν are Slater determinants of hydrogenlike orbitals sharing one weighted charge $Q_\nu = p_\kappa / R_\nu$, with configuration root $R_\nu = \sqrt{\sum_j 1/n_j^2}$, all isoenergetic at $E = -\frac{1}{2}p_\kappa^2$. The kinetic term cancels against V_0 by construction, the nuclear-attraction matrix is diagonal, $T^0_{\nu'\nu} = ZR_\nu\,\delta_{\nu'\nu}$, and the atomic secular equation is [3, 4]

$$\sum_\nu \Big[ZR_\nu\,\delta_{\nu'\nu} + T'_{\nu'\nu} - p_\kappa\,\delta_{\nu'\nu} \Big] B_\nu = 0, \qquad E = -\tfrac{1}{2}p_\kappa^2. \tag{3}$$

Three features of Eq. (3) are load-bearing for a quantum algorithm, and all are consequences of the potential-weighting:

1. It is a *standard* eigenproblem, $MB = p_\kappa B$ with $M = \mathrm{diag}(ZR_\nu) + T'$. There is *no overlap metric S*: the potential-weighting has absorbed it into the diagonal nuclear term. The ill-conditioning of Sec. II is not repaired, it is *never present*.

2. Its eigenvalues are the scaling parameters p_κ *directly*; one diagonalization yields the whole spectrum via $E = -\frac{1}{2}p_\kappa^2$. There is no self-consistency and no outer search over the energy.

3. The interelectron matrix $T'_{\nu'\nu} = -p_\kappa^{-1}\int \Phi^*_{\nu'} V' \Phi_\nu$ is a matrix of *pure numbers*, independent of p_κ, of E, and of Z: the two-electron integrals scale as $Q_\nu = p_\kappa/R_\nu$, and the $-p_\kappa^{-1}$ prefactor cancels the p_κ. A single T' serves an entire isoelectronic series.

T' is assembled by Slater–Condon reduction of the two-electron operator, with the $1/r_{12}$ Laplace expansion giving a finite multipole sum (the angular factors are products of Gaunt integrals / Wigner $3j$, recoupling to $6j$ for coupled configurations) times a radial Slater integral [4].

IV. ATOMIC IMPLEMENTATION, VALIDATION, AND THE SUBLINEAR 1-NORM

We implement Eq. (3) for helium ($Z = 2$). For the leading configuration $1s^2$ the matrix is 1×1: $R_{1s^2} = \sqrt{2}$, the unit-charge integral is $\langle 1s^2 | r_{12}^{-1} | 1s^2 \rangle = \frac{5}{8}$, so

$$T'_{1s^2,1s^2} = -\tfrac{5}{8}/\sqrt{2} \approx -0.442, \quad p_\kappa = Z\sqrt{2} + T' = 2.386, \tag{4}$$

giving $E = -p_\kappa^2/2 = -2.847$ Ha. **[MEASURED]** Computed on a radial grid this reproduces the textbook variational helium value (-2.84766 Ha; the pure number $\frac{5}{8}/\sqrt{2}$ is confirmed to grid precision), and the "bare" matrix (no V') returns -4.0 Ha, i.e. two He^+ $1s$ orbitals—the exact non-interacting limit. **[MEASURED]** Adding configurations coupled to total ^1S lowers the energy monotonically,

$$-2.847\,(1s^2) \;\to\; -2.873\,(s) \;\to\; -2.894\,(+p) \;\to\; -2.897\,(spdf,\ K{=}164),$$

converging toward the exact non-relativistic -2.90372 Ha. **[MEASURED]** The residual ~ 7 mHa is genuine basis incompleteness, not a missing metric: the Goscinskian basis is deliberately poor for the helium ground state—Avery and Avery reach -2.90250 with 102 optimized Coulomb–Sturmian configurations [2], still 1.2 mHa short—and every point stays above the exact value (no variational overshoot). The machinery is correct, including the higher-ℓ angular correlation (Gaunt / Wigner-$3j$ multipole coupling).

The decisive quantity is the block-encoding cost. **[MEASURED]** The entrywise 1-norm of the secular matrix, $\|M\|_1 = \sum_{\nu'\nu} |M_{\nu'\nu}|$ (an upper bound on the achievable LCU λ), grows *sublinearly* in the number of configurations K,

$$\|M\|_1 \sim K^{0.84} \quad \text{(full } s{+}p{+}d{+}f \text{ basis)}, \tag{5}$$

the opposite of the naive inflation Eq. (1). The mechanism is the one Avery notes for Coulomb potentials: the pure-number matrix elements decrease with quantum number (higher configurations have small root R_ν and rapidly decaying couplings), so the sum grows more slowly than the matrix dimension. Crucially, angular correlation does *not* break it: the s-only exponent 0.78 rises only to 0.84 with p, d, f configurations present, so the contrast with the naive superlinear inflation holds for the full basis, not merely the s-sector. Two caveats remain: the construction uses L^2-normalized configurations rather than the exact potential-weighted normalization (the single-configuration value is exact and the convergence is correct, so the constant may shift but not, we expect, the sublinear exponent); and K is the matrix dimension, not the qubit count, so the robust statement is the *qualitative* contrast, sublinear versus the L^2 superlinear blow-up. The quoted exponents are numerical fits at the computed basis sizes; the load-bearing claim is that regime and ordering, not the precise fitted value.

V. THE QUANTUM ALGORITHM AND ITS NOVELTY

Equations (3)–(5) describe an unusually clean target for quantum eigenvalue estimation: block-encode the fixed, bounded, pure-number matrix M; estimate its eigenvalues p_κ by qubitization or phase estimation [9]; return the energies $E = -\frac{1}{2}p_\kappa^2$. **[OBSERVATION]** A targeted prior-art search finds no existing quantum algorithm combining (i) a Sturmian/hyperspherical basis, (ii) a quantum eigenvalue routine, and (iii) the isoenergetic inversion (fix E, solve for a scale). Coulomb–Sturmian bases are classical in the literature [5]; quantum electronic structure uses Gaussian and plane-wave bases; and quantum generalized-eigenvalue solvers [10, 11] run the forward direction. The closest relative [11] scans a candidate eigenvalue as a singularity test, which is not the isoenergetic advantage. The atomic algorithm therefore appears novel.

The two documented cost risks for such a scheme are, moreover, *absent in the atomic case*. The metric-conditioning multiplier of block-encoded generalized eigenproblems [10, 12] does not apply, because Eq. (3) carries no metric. The outer energy search that would otherwise multiply the cost does not arise, because the eigenvalues are the energies. What remains is a single figure of merit, the 1-norm of M, which Eq. (5) measures to be sublinear. We stress that this is a resource analysis, not a hardware demonstration.

VI. THE MOLECULAR FRONTIER: SHIBUYA–WULFMAN CONDITIONING

The clean atomic result does not fully transfer to molecules. The multi-center one-electron isoenergetic problem is again *generalized*,

$$\sum_{\mu} \left[W_{\mu'\mu} - k\, S_{\mu'\mu} \right] C_\mu = 0, \tag{6}$$

with S the Shibuya–Wulfman matrix [3], whose condition number multiplies the block-encoding cost: inverting or square-rooting S by quantum singular-value transformation costs a polynomial degree growing with κ (Sec. VI A) [13, 14]. In Hermitian form $S_{\mu'\mu} = (2k^2)^{-1}\langle\nabla\varphi_{\mu'}|\nabla\varphi_\mu\rangle + \frac{1}{2}\langle\varphi_{\mu'}|\varphi_\mu\rangle$.

[MEASURED] On a two-center s-orbital Coulomb–Sturmian basis we find: (i) the intra-center block of S is *exactly* the identity—in the momentum-space evaluation, diagonal $\{1, 1, 1\}$ with off-diagonal $\sim 10^{-16}$ (the 2D-grid values $\{1, 0.999, 0.998\}$ are discretization error)—where the L^2 overlap's intra-center block is ill-conditioned (cond 5.8, itself growing with basis), so the within-center disaster of Sec. II is genuinely absent from S; (ii) across two centers S is uniformly better-conditioned than the L^2 overlap and markedly better at larger separation:

R (bohr)	basis	cond(L^2)	cond(S)
1.4	6	39.1	30.1
2.0	6	20.9	15.4
4.0	6	11.1	**4.4**

but (iii) cond(S) still grows with basis size—*polynomially*, cond(S) $\sim N^{1.85}$, essentially the same rate as the L^2 overlap's $N^{1.70}$ (decisively power-law, not exponential), with the largest eigenvalue bounded, $\lambda_{\max} < 2$, and all the growth in $\lambda_{\min} \to 0$ as the basis approaches linear dependence—driven by the inter-center coupling (off-block magnitude $\sim$ 0.4–0.6), so the improvement over L^2 is only modest at bonding distance. All three cross-check across a 2D position-space grid, the closed-form momentum-space quadrature, and a direct three-dimensional momentum integral (agreement to $\sim 10^{-10}$).

[SYMBOLIC + MEASURED] The conditioning law is in fact *derived*, and the fitted exponent is a window reading. Because the intra-center block is exactly the identity, $S = \left[\begin{smallmatrix} I & C \\ C^{\mathsf{T}} & I \end{smallmatrix}\right]$ has spectrum $\{1 \pm \sigma_k\}$ with σ_k the singular values of the cross block (the cosines of the principal angles between the two center subspaces), so $\mathrm{cond}(S) = (1+\sigma_{\max})/(1-\sigma_{\max})$ *exactly.* In the sine basis the cross block is the finite section of a multiplication operator with symbol $j_0(kR\cot(\chi/2))$, whose supremum 1 is attained at $\chi = \pi$; the band-limited concentration rate at a quadratic symbol maximum gives

$$1 - \sigma_{\max} \;=\; \frac{(kR)^2}{24}\,\frac{\pi^2}{n^2} + o(n^{-2}), \tag{7}$$

(n orbitals per center), i.e. the conditioning exponent is exactly 2 asymptotically, with the single-parameter collapse $(1 - \sigma_{\max})\,(n/kR)^2 \to \pi^2/24$ (confirmed to $\sim 1\%$ at $n = 160$; the leading-order derivation is asymptotic, the remainder is measured, not bounded). The fitted $N^{1.85}$ above—and the $N^{1.81}$ of the same quantity at $R = 2$ in the sprint record—are pre-asymptotic slopes of this one law read on different windows. The same singular spectrum carries the composition wall of the balanced-basis arc: the center-projection commutator is $\|[P_A, P_B]\| = \max_k \sigma_k \sqrt{1 - \sigma_k^2}$, which *saturates* at its algebraic ceiling $\frac{1}{2}$ (some σ_k lands near $1/\sqrt{2}$) as the spectrum densifies—the previously reported 0.50 is the saturation value—so the two walls are functionals of one object, with all of the severity information in $1 - \sigma_{\max}$ (`tests/test_paper60_sigma_law.py`).

The molecular metric cost is therefore *mitigated, not dissolved*: Eq. (6) retains a polynomial conditioning multiplier that is worst where the chemistry lives, and it cannot simply be transformed away—conditioning is orthogonal-invariant, so the momentum-space form (which cancels the Fock Jacobian to a clean one-dimensional quadrature) yields the *identical* matrix and is an exact, grid-free *evaluation* win, not a conditioning improvement. Two genuine conditioning levers do exist: at larger separation cond(S) $\to$ 1 rapidly while the L^2 overlap cannot follow (its intra-center conditioning grows with basis regardless); and gerade/ungerade symmetry adaptation exactly block-diagonalizes S, isolating a perfectly-conditioned gerade sector (cond ≈ 2, flat in basis and R) and confining all ill-conditioning to the ungerade sector—a free $\sim 2\times$ effective-cost reduction in the homonuclear case. The gerade flatness is itself derived: for equivalent centers the blocks are exactly $I \pm C$, and the near-dependent direction has symbol value $+1$, so it lives entirely in $I - C$ (ungerade) while $\mathrm{cond}(I + C) \to 2/(1 + \min_x j_0(x)) = 2.555041\ldots$, an exact constant of the sinc symbol independent of R and of basis size (`tests/test_paper60_sigma_law.py`); the "$\kappa \approx 2$" rows below are this constant read at small n.

[MEASURED] This lever is a property of *equivalent* centers, not of symmetry as such, and a probe on water (H_2O, C_{2v}: O + 2H) makes the distinction sharp. For an s-only shared-scale basis the only symmetry action is the H $\leftrightarrow$ H swap (O sits on the C_2 axis, fixed), so the totally-symmetric A_1 block—which holds the ground state—contains *both* the O functions and the symmetric H combinations, and with them the O $\leftrightarrow$ H coupling between symmetry-*inequivalent* centers that the group cannot remove. Building water's three-center SW metric exactly (every s–s block depends only on inter-center distance) and adapting to C_{2v}, the A_1 ground-state block's conditioning *grows*, cond(A_1) $\sim N^{1.97}$ ($R^2 = 1.000$), essentially the raw metric ($19.9 \to 698$ over $N = 6 \to 36$)—*not* flat like the H_2^+ gerade sector (cond ≈ 2 throughout). Zeroing only the O $\leftrightarrow$ H block collapses it back to cond ≈ 2, pinning that inequivalent-center coupling as the sole driver. The lever therefore does not transfer to a polyatomic with a symmetry-unique heavy center; it is confined to homonuclear-diatomic (equivalent-center) systems and their close analogs. The water block obeys the same derived law: the O $\leftrightarrow$ (H + H) canonical correlations follow Eq. (7) with symbol curvature $c_{\rm sym} = R_{\rm OH}^2/24 - R_{\rm HH}^2/96$ (exponent again exactly 2; the fitted 1.97 is the same pre-asymptotic reading), and $\mathrm{cond}(A_1) \approx 1.12 \times (1 + \sigma_{\max})/(1 - \sigma_{\max})$ with a constant prefactor—the canonical correlation is the whole story. The structural reason the symmetry cannot help is now exact: O and H are not exchanged by any group element, so both $1 \pm \sigma$ eigenvector families remain inside A_1 and no block absorbs the divergent factor. A low-rank escape is also closed *structurally*: the coupling's singular spectrum discretizes the range of the symbol (its effective rank grows linearly with basis size, participation ratio ≈ 7 at $N = 36$), and since $1 - \sigma_k \approx c_{\rm sym}(k\pi)^2/n^2$, an exact rank-$r$ (Woodbury) treatment of the top correlations divides the residual κ by $(r+1)^2$ only—a useful constant factor (≈ 40–$75\times$ in $d_{\rm inv}$ terms at $N = 12$–36) but never a change of exponent; flattening would need $r \propto n$, i.e. classically diagonalizing the whole coupling, the sparsity-destroying orthogonalization already excluded.

A. Resource estimate: the gerade lever is the molecular payoff

We convert the measured conditioning into a block-encoding resource count for the one-electron H_2^+ secular equation Eq. (6), at chemical accuracy ($\epsilon_E = 1.6$ mHa) and $R = 2$ bohr. We whiten the generalized problem to a *standard* eigenproblem $\tilde{M} = S^{-1/2} W S^{-1/2}$ whose eigenvalues are the scales k; the metric enters only through $S^{-1/2}$, applied by quantum singular-value transformation with a polynomial approximating $x^{-1/2}$ on $[\kappa^{-1}, 1]$ of degree $d_{\rm inv} \sim \kappa \ln(\kappa/\epsilon)$ [13, 14], $\kappa = \mathrm{cond}(S)$. This $d_{\rm inv}$ *is* the metric penalty—the quantitative content of the statement that the conditioning multiplies the block-encoding cost [13, 14]. Eigenvalue estimation to precision $\epsilon_k = \epsilon_E/k$ costs $Q \sim (\pi/2)\lambda_{\rm eff}/\epsilon_k$ qubitization queries [9], with subnormalization $\lambda_{\rm eff} = k_{\max}$; each query calls one block-encoding of W and $d_{\rm inv}$ of S, so the metric multiplies the query count by $(1 + d_{\rm inv})$.

[MEASURED] The algorithm binds: solving Eq. (6) for H_2^+ by isoenergetic self-consistency (build the basis at scale k, solve $[W - k'S]C = 0$, locate $k' = k$) gives the σ_g ground root $k = 1.475$, i.e. $E_{\rm elec} = -1.088$ Ha against the reference -1.103 Ha—a 1.3% s-only underbinding (no p-polarization, the expected basis limitation, above the exact value)—and pins $\lambda_{\rm eff} = k_{\max} \approx 1.7$.

[RESOURCE MODEL, on the measured κ] Table I gives the counts. Three readings: (i) the metric-free atomic case ($\kappa = 1$, $d_{\rm inv} = 0$) is recovered as the zero-penalty limit; (ii) the full SW metric costs $d_{\rm inv} \sim 10^3$ by $N = 16$; but (iii) the H_2^+ ground state σ_g is *gerade*, and the gerade sector's conditioning is flat ($\kappa_g \approx 2$ across $N = 4$–20), so the ground-state $S^{-1/2}$ costs $d_{\rm inv} \approx 16$–18 *independent of basis size*—a $\sim 60\times$ reduction versus the full metric at $N = 16$, on a register halved to $\lceil \log_2(N/2) \rceil$ qubits. The gerade lever is thus not merely the $2\times$ symmetry factor of the previous paragraph: it delivers the chemically relevant state at a metric penalty that *does not grow with basis size*. This is the molecular payoff.

TABLE I. Block-encoding resource for the one-electron H_2^+ isoenergetic secular equation Eq. (6) at chemical accuracy ($\epsilon_E = 1.6$ mHa, $R = 2$ bohr), $N = 16$ ($n_{\max} = 8$ per center). $d_{\rm inv}$ is the QSVT degree for $S^{-1/2}$ (the metric penalty); qubits $= \lceil \log_2 N_{\rm dim} \rceil + n_{\rm phase} + n_{\rm anc}$ with $n_{\rm phase} = 10$. The qubitization query count $Q \approx 2.5 \times 10^3$ is common to every row (it depends on $\lambda_{\rm eff}$ and ϵ_k, not the metric). A resource model on the measured κ; the $O(1)$ constant in $d_{\rm inv}$ cancels in the cross-metric ratios but not in the absolute counts.

metric	κ	$d_{\rm inv}$	qubits
atomic (no metric, Sec. III)	1	0	23
SW gerade (σ_g ground)	2.3	18	26
SW full	91.8	1050	31
Gaussian LCAO (ratio 2)	1.99×10^4	3.3×10^5	31

[RESOURCE MODEL] Against the same H_2^+ problem in a standard even-tempered s-Gaussian LCAO metric (ratio 2), the SW inverse-degree is 10^2–3×10^2 times smaller (full metric) and 10^3–10^4 smaller (gerade). This is a metric-versus-metric comparison and carries an honest caveat: the Gaussian conditioning is ratio-dependent—a denser ratio-1.6 set (approaching completeness) blows up as cond $\sim N^6$ ($\kappa \sim 10^5$ at $N = 16$), a sparser ratio-3 set grows only as $N^{1.4}$ but leaves coverage gaps and still carries $\sim 10\times$ the SW prefactor. This is the classic Gaussian coverage-versus-linear-dependence trade-off; the Coulomb–Sturmians evade it because they are a *complete* set at one scale whose metric is polynomially conditioned ($\mathrm{cond}(S) \sim N^{1.85}$ on this window; derived asymptote N^2, Eq. (7)) with a small prefactor. Production Gaussian simulation of course absorbs the overlap classically in a prior mean-field step; the metric penalty priced here is precisely the cost of the on-device angular generation that motivated the Sturmian encoding in the first place.

[MEASURED] The large-R lever is quantified alongside: $\mathrm{cond}(S)$ falls to 2.2 at $R = 10$ bohr (at the sweep's $N = 12$, $d_{\rm inv}$: $862 \to 17$ over $R = 1.4 \to 10$; at Table I's $N = 16$ the $R = 1.4$ figure is larger still, so the lever is stronger than these numbers suggest), while the Gaussian metric stays $\sim 2 \times 10^3$—its intra-center linear dependence does not relax with separation.

[MEASURED] Two probes price the derived law's exploitability (resource-model tier; CHANGELOG v4.105.0). *Spectrum-aware inversion*—replacing the worst-case interval approximation for $S^{-1/2}$ by constrained interpolation on the *known* spectrum $\{1 \pm \sigma_k\}$ (classically computable by an $O(N^3)$ values-only SVD that touches no integral sparsity; the asymptotic law alone is 11–21% off at $\lambda_{\min}$ and does not suffice)—buys a constant factor $\approx$ 8–10× at the *same* κ exponent, and the Bernstein $\Theta(\kappa)$ floor for *both* problems bounds it there (an argument, not a measurement: the floors hold for the interpolation and interval families alike, because the small eigenvalues are spaced at exactly the scale on which $x^{-1/2}$ varies). At this table's $N = 16$ point the honest chain is model 1050 → same-convention minimax 209 → aware 27. An *exponent-changing* variant exists—a PSD shift of $[\lambda_{\min}, \lambda_{\max}]$ onto $[-1, 1]$, needing only Eq. (7)'s closed-form endpoints, gives degree $\approx 2.8\sqrt{\kappa}$—but is gated by its LCU subnormalization (the naive 1-norm ≈ 3 squeezes the spectrum back to linear-in-κ, *worse* than baseline; with $\sim 3\times$ amplification the net at $\kappa = 864$ is $\approx 5 \times 10^2$ effective queries vs $\approx 1.9 \times 10^3$)—scoped, not priced beyond this. Separately, an *analytic momentum-space factorization* of the ERI tensor (the closed-form momentum densities as a continuous factorization index) matches but does not beat plain density fitting: eigen-Cholesky DF on the same tensor reaches within 13% of its 1-norm at rank $\leq n_{\text{orb}}^2$, versus 2.5–4.3 × 10^4 momentum leaves—the observed gains are DF's, not the momentum representation's.

B. Many-electron molecules: the metric does not compound

For a many-electron molecule the generalized-Sturmian secular equation is still $[T^0 + T' - p_\kappa \not\Vdash]B = 0$—a *standard* eigenproblem, the $-p_\kappa \not\Vdash$ being the identity, not a metric (Avery's general form [3]); only the atomic conveniences "T^0 diagonal" and "T' pure numbers" are lost. The resource-relevant question is whether the Shibuya–Wulfman metric penalty of Sec. VI *compounds* with electron number. It need not.

[SYMBOLIC + MEASURED] A k-electron configuration is a Slater determinant of one-electron molecular orbitals, so its configuration-space overlap is the k-th compound matrix of the one-electron overlap, with condition number $(\lambda_1 \cdots \lambda_k)/(\lambda_{N-k+1} \cdots \lambda_N)$ in the one-electron overlap spectrum $\{\lambda_i\}$. Built from the *raw* L^2-normalized many-center Coulomb–Sturmians this *grows* with electron number—on the H_2-geometry two-center basis ($N = 6$), $1\times, 8\times, 9 \times \operatorname{cond}(S_{L^2})$ at $k = 1, 2, 3$ (worst at half-filling), bounded by $\operatorname{cond}(S_{L^2})^{\min(k,N-k)}$ and driven by the near-linear-dependence eigenvalue. Built instead from the *molecular Sturmians*—the Shibuya–Wulfman eigenvectors, S_{SW}-orthonormal by construction—the configuration overlap is the *identity at every electron number*—an algebraic consequence of the compound-matrix identity, not a numerical finding; only the $1\times, 8\times, 9\times$ L^2 figures above are measured. The metric is therefore paid once, at the one-electron molecular-orbital construction ($\operatorname{cond}(S_{\text{SW}})$ above, with the gerade lever), and the N-electron secular equation inherits an identity metric: **the molecular metric penalty is one-electron, not N-electron**. It is the on-device analogue of the once-executed classical mean-field step that lets production Gaussian simulation carry no metric into the many-body Hamiltonian.

[MEASURED] The interelectron matrix T' is assembled from two-center electron-repulsion integrals, and these are in hand. We evaluate the shared-scale Coulomb–Sturmian two-center ERIs $(pq|rs)$ numerically (a single multipole route about one nucleus, so every center class is uniform) and validate them against the framework's *exact* closed form: the one-center $(1s\,1s|1s\,1s) = \frac{5}{8}$ to 4×10^{-6}, and the two-center $(AA|BB)$ to $\sim 10^{-4}$ against `aabb_value` at $R \in \{1.4, 2.0, 3.0\}$ (the closed forms themselves are weight-one, π-free $\{E_1, \ln, \gamma\}$ [15, 16]). With these, the interacting method binds a molecule: minimal-basis ($1s$ on each center, common scale) H_2 in the metric-free molecular-Sturmian basis has an interior minimum at $R \approx 1.6$ bohr with $E_{\text{tot}} \approx -1.09$ Ha—the single-ζ minimal-basis value, underbinding the exact -1.174 Ha as s-only single-ζ must, above the exact energy throughout.

[OPEN, with the obstruction identified] What remains is the block-encoding 1-norm of the full N-electron secular matrix $T^0 + T'$, and here the molecular case parts from the atomic one at a structural level, not merely a computational one. The atomic sublinearity of Eq. (5) rides on the *clean diagonal* $T^0 = ZR_\nu$, which exists only because (i) the Goscinskian orbitals scale *with* p_κ (charge $Q_\nu = p_\kappa/R_\nu$), so the p_κ in $\langle 1/r \rangle$ cancels the $-p_\kappa^{-1}$ prefactor and M is a p_κ-independent linear matrix, and (ii) on a *single* center $\langle 1/r \rangle = Q/n^2$ collapses the collective scale to the root-sum-of-squares $T^0_{\text{bare}} = \sqrt{\sum_p k_p^2} = ZR_\nu$ (we verify: for He $1s^2$, $\sqrt{k^2 + k^2} = 2\sqrt{2} = ZR_\nu$ gives the exact bare -4 Ha; a naive sum of scales gives -8). On *two* centers the nuclear attraction $\langle 1/r_A \rangle + \langle 1/r_B \rangle$ acquires off-center terms depending on $Q_\nu R$, so T^0 is neither diagonal nor p_κ-independent, and a fixed-scale molecular-Sturmian shortcut is not the isoenergetic matrix at all—it fails the non-interacting limit (returning $p_\kappa = 2k_{\sigma_g}$ where the correct value is $\sqrt{2}\,k_{\sigma_g}$). A faithful molecular 1-norm therefore requires the p_κ-scaled molecular Goscinskian (config charge $Q_\nu = p_\kappa/R_\nu$, two-center nuclear T^0 and two-center ERIs at the config scale, solved self-consistently; validated at $R \to \infty = 2$ H atoms and $R \to 0 =$ He). The obstruction is thus not that the integrals are hard—they close (Papers [15, 16])—but that the single-center diagonal-T^0 structure that *produced* the sublinearity does not transfer.

[MEASURED] We can nonetheless price the practical alternative. Building the two-center mixed-charge integrals (overlap, nuclear attraction, and ERIs, validated against the exact closed forms and against $\langle 1s|1/r_B|1s \rangle$ to $\sim 10^{-4}$) and a full two-electron CI in the Loewdin-orthonormalized Sturmian basis yields a *validated* molecular solver—H_2 dissociates correctly to two H atoms ($E_{\text{tot}} \to -1.0$ Ha at $R = 6$) and binds above the exact -1.174 Ha at R_{eq}. Its *standard* block-encoding 1-norm $\lambda = \sum_{pq} |h_{pq}| + \sum_{pqrs} |(pq|rs)|$ grows polynomially, $\lambda \sim n_{\text{orb}}^{2.2}$, with no sublinear behavior—the atomic advantage of Eq. (5) is a property of the isoenergetic secular matrix, not of the basis, and it does not survive the loss of the diagonal T^0. For molecules the resource question therefore answers in the ordinary way: the block-encoding cost scales polynomially, and the metric levers of Secs. VI A–VI B (the gerade sector, large-R, one-electron-only metric) are the real savings, not a sublinear matrix.

VII. CONCLUSION AND SCOPE

The message is a sharp split. *For atoms*, the isoenergetic generalized-Sturmian reformulation converts electronic structure into a standard, metric-free eigenproblem with a bounded pure-number matrix whose block-encoding 1-norm grows sublinearly—a clean, apparently novel quantum secular equation in which the two feared cost risks of the genre are structurally absent. The L^2-encoding obstruction that makes the same basis the worst possible choice is revealed as an artifact of the wrong framing. *For molecules*, the Shibuya–Wulfman metric returns; it is genuinely better-conditioned than the L^2 overlap—its intra-center block is the identity—and much better at large separation, but its inter-center coupling still costs conditioning that grows with basis size. The one qualification that lifts this from a caveat to a payoff is symmetry: because the H_2^+ ground state is gerade and the gerade sector is perfectly conditioned and flat in basis size, the ground-state block-encoding pays a metric penalty that does not grow (Table I)—the growth is confined to the excited/ungerade states—though this is an *equivalent*-center property: a probe on water (C_{2v}) shows a symmetry-unique heavy center reinstates the growth in the ground-state (A_1) block (Sec. VI), confining the lever to homonuclear-diatomic-like systems. And the penalty does not compound with electron number: the Shibuya–Wulfman metric is discharged once, at the one-electron molecular-orbital construction, after which the many-electron secular equation carries an identity metric (Sec. VI B)—the metric cost is one-electron, not N-electron. The atomic case is a result; the molecular case is a frontier with a well-conditioned ground-state corner.

We have not run the algorithm on hardware, and we do not claim a resource advantage over production quantum chemistry: the systems where the encoding is demonstrably cheap are small. What we claim is structural and measured—that the generalized-Sturmian method's classical elegance (no self-consistency, energies direct, pure-number matrices) is also, for atoms, a near-ideal quantum encoding—and that this reframes the on-device-generation intuition from a hope into a characterized boundary, clean on one side and open on the other.

ACKNOWLEDGMENTS

The generalized-Sturmian formulation used here is that of J. E. Avery and J. S. Avery; this paper adds only its reading as a quantum-encoding target and the resource measurements. Developed with an AI-augmented workflow under principal-investigator direction, with the headline numerical claims validated against analytical benchmarks and a regression test suite.

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